Computational Finance - Lab Assignment 3

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1 PDE Methods: Binary and Barrier Options

In this assignment, you will implement and analyze numerical methods for pricing two classes of exotic options:

(a) **Binary (Digital) Options**

(b) **Barrier Options (Up-and-Out, Down-and-In)**

You will use Monte Carlo, closed-form formulas under Black–Scholes, and a finite-difference PDE approach (implicit scheme).

Binary options

Consider a price process $(S_t)_{t\in\mathbb{R}^+}$ given by

$$\frac{dS_t}{S_t} = rdt + \sigma dB_t, \quad S_0 = 1 \tag{1}$$

under the risk-neutral probability measure $\mathbb{Q}$. The binary (or digital) call option is a contract with maturity T, strike price K and payoff

$$C_d := \mathbb{1}_{[K,\infty)}(S_T) = \begin{cases} 1\ Eur & S_T \geq K \\ 0\ Eur & S_T < K \end{cases} \tag{2}$$

1. Derive the Black-Scholes PDE satisfied by the pricing function $C_d(t, S_t)$ of the binary call option, together with its terminal condition.
2. Show that the solution of $C_d(t, x)$ of the Black-Scholes PDE is given by

$$\begin{aligned} C_d(t,x) &= e^{(T-t)r}\Phi\left(\frac{(r-\sigma^2/2)(T-t)+\log(x/K)}{\mid\sigma\mid\sqrt{T-t}}\right) \\ &= e^{-(T-t)r}\Phi(d_-(T-t)) \end{aligned}$$

   where

$$d_-(T-t) := \frac{(r-\sigma^2/2)(T-t)+\log(x/K)}{\mid\sigma\mid\sqrt{T-t}}, \quad 0 \leq t < T \tag{3}$$

3. Use Monte carlo to compute the price of a binary option to its closed form solution.
4. Use the implicit discretization scheme and the Crank Nicolson scheme to price a binary/digital options. Make also a sensitivity analysis on the parameter space and plot the digital option surface $c(S,t)$. Finally compute the delta as a function of the underlying S.

Knock-out Barrier options

We are now going to consider a slightly more exotic option, the so called **Knock-out barrier** options. Let us consider an up-and-out barrier call option with maturity T, strike price K barrier (or call level B) whose payoff is given by

$$C = (S_T - K)\,\mathbb{1}_{\max_{0\leq t\leq T} S_t < B} \tag{4}$$

with $B \geq K$ so that the payoff is not zero. In this case the option expires worthless if the barrier is not triggered. If instead the barrier is hit the option becomes a vanilla option.
The goal this exercise is to compute the fair value of such path-dependent options using a closed form formula, Monte carlo methodologies and the BS PDE with an appropriate terminal condition.

1. Show that when $K \leq B$ the price

$$e^{-(T-t)r}\mathbb{1}_{\{M_0^t<B\}}\mathbb{E}\left[\left(x\frac{S_{T-t}}{S_0}-K\right)^+\mathbb{1}_{\{x\max_{0\leq u\leq T-t}\frac{S_u}{S_0}<B\}}\right]_{x=S_t} \tag{5}$$

of the up-and-out barrier call option with maturity T strike price K and barrier level B is given by

$$\begin{aligned}
C(t,T,S_T,M_0^T) &:= C(H)\\
&= e^{-(T-t)r}\mathbb{E}\left[(S_T-K)^+\mathbb{1}_{\{M_0^T<B\}}\mid\mathcal{F}_t\right]\\
&= S_t\mathbb{1}_{\{M_0^t<B\}}\{\Phi\left(\delta_+^{T-t}\left(\frac{S_t}{K}\right)\right)-\Phi\left(\delta_+^{T-t}\left(\frac{S_t}{B}\right)\right)\}\\
&- S_t\mathbb{1}_{\{M_0^t<B\}}\left(\frac{B}{S_t}\right)^{1+2r/\sigma^2}\left(\Phi\left(\delta_+^{T-t}\left(\frac{B^2}{KS_t}\right)\right)-\Phi\left(\delta_+^{T-t}\left(\frac{B}{S_t}\right)\right)\right)\\
&- e^{-r(T-t)}K\mathbb{1}_{\{M_0^t<B\}}\{\Phi\left(\delta_-^{T-t}\left(\frac{S_t}{K}\right)\right)-\Phi\left(\delta_-^{T-t}\left(\frac{S_t}{B}\right)\right)\}\\
&+ \left(\frac{S_t}{B}\right)^{1-\frac{2r}{\sigma^2}}e^{-r(T-t)}K\mathbb{1}_{\{M_0^t<B\}}\{\Phi\left(\delta_-^{T-t}\left(\frac{B^2}{KS_t}\right)\right)-\Phi\left(\delta_-^{T-t}\left(\frac{B}{S_t}\right)\right)\}
\end{aligned}$$

where

$$\delta_{\pm}^{\tau}(z)=\frac{1}{\sigma\sqrt{\tau}}\left(\log z+\left(r\pm\frac{\sigma^2}{2}\right)\tau\right),\quad z>0 \tag{6}$$

2. Use the analytical formula above to compare it with the price you derive using a MC simulation. Compare the values of the option wrt to the option parameters as well as the model's parameters. For the MC simulation we are going to use an adjusted/corrected methodology proposed in the following paper (Adjusted MC price barrier option). If we define by $C(H)$ the price of the above continuous up and out call option and then we define by $C_m(H)$ the price of an otherwise identical discrete barrier option (for example the one we derive from a MC simulation). The we have that $C_m(H)=C(He^{\beta_1\sigma\sqrt{T/m}})+o(\frac{1}{\sqrt{m}})$. Here $\beta_1\approx-\frac{\zeta(1/2)}{\sqrt{2\pi}}\approx 0.5826$ with ζ the Rienmann zeta function. Run also a convergence analysis on the MC method and comment on the convergence rates.[1]

3. Use the implicit discretization scheme to price the barrier up and out call option. Make also a sensitivity analysis on the parameter space and plot the barrier option surface $c(S,t)$. Finally compute the delta as a function of the underlying S.

[1] One sees this result as saying that to price a discretely monitored barrier option using the continuous formula one shoudl first shift the barrier away from S_0 by a factor $e^{\beta_1\sigma\sqrt{T/m}}$. This corrects for the fact that when the discrete time process $\{S_{k\Delta},k=0,1,...\}$ breaches the barrier it overshoots. The constant $\beta_1\sigma\sqrt{T/m}$ should be viewed as an epproximation to the overshoot in the logarithm of the price of the underlying.

2 Assignment 2 - Hedging with Real SPX Option Data (SABR and Bartlett Deltas)

In this assignment you will implement and evaluate **one-step (one-day) delta hedging strategies** for SPX index **European call options** using real market data. This assignment is a natural extension of Assignment 1: you will retain the same empirical hedging pipeline developed there for real SPX option data, but move beyond the flat-volatility Black–Scholes benchmark to a smile-aware stochastic-volatility specification. The theoretical motivation for this step is discussed in [1], where Bartlett's delta is interpreted through the lens of *variance-optimal hedging*: when implied volatility evolves dynamically, the hedge that minimizes mean-square hedging error is no longer the plain delta hedge, but a delta corrected by a volatility-covariation term scaled by vega. In the lognormal SABR model this variance-optimal correction coincides with Bartlett's delta, and it can also be shown that its advantage over standard delta hedging becomes more pronounced when the leverage correlation ρ is large in absolute value. Thus, this assignment connects your implementation from Assignment 1 with both a more realistic smile model and the theoretical idea of variance-optimal hedging under lognormal SABR.

You will extend the Black–Scholes (BS) baseline hedge which you implemented in the first assignment by:

(i) calibrating a **lognormal SABR** smile ($\beta = 1$) to market implied volatilities,

(ii) computing the **SABR delta** and the **Bartlett delta** (skew-adjusted delta),

(iii) and comparing hedging effectiveness using **SSE/MSE** and a standardized **Gain** metric.

Competition. This assignment is **competitive**: you must report results using the *exact* standardized procedure below (filters, bins, definitions, and file formats) so that results are comparable across students and can be scored automatically.

Learning objectives

By completing this lab you will:

- implement one-step (one-day) delta hedging using real option data (this should already have been implemented in the first assignment),
- calibrate a **lognormal SABR** volatility smile ($\beta = 1$, shift $\theta = 0$) from implied volatilities,
- compute **SABR delta** (smile-aware chain rule) and **Bartlett delta** (skew adjustment),
- compare hedging performance using **SSE**, **MSE**, and the **Gain** metric,
- produce **standardized CSV outputs** for competition scoring.

Dataset

The dataset consists of SPX call option quotes over the period **2023-02-01 to 2023-02-28** (WRDS export). For each trading date t and each option contract (strike K, expiry T), the dataset contains at least:

(`date`, `exdate`, `symbol`, `strike_price`, `best_bid`, `best_offer`, `impl_volatility`, `delta`, `cp_flag`).

You must use `cp_flag = 'C'` (calls only).

Underlying SPX close series. Let S_t denote the SPX close on date t. If S_t is not included in the WRDS export, you must source it externally and merge it by date. **Important:** since the evaluation uses one-day changes $\Delta S_t = S_{t+1} - S_t$, you must obtain *at least one additional trading day after 2023-02-28.*

Interest rates and dividends. For competition scoring you must use the *same* convention everywhere:

- Either set $r = 0$ and $q = 0$ everywhere, *or* (recommended) use the provided/starter procedure to attach a daily risk-free zero rate $r_t(\tau)$ and a dividend-yield proxy q_t.
- If you include dividends, your hedge move must be the one-step *total return* move $\Delta S_t^{TR} := \Delta S_t + S_t\big(e^{q_t \Delta t} - 1\big)$.
- Whichever convention you choose, it must be used *consistently* for: forward mapping F_t, BS deltas, SABR calibration, and hedging residuals.

Structure and scoring

This assignment has three graded parts:

- **Chunk A (baseline BS hedging):** construct your own BS delta and compute baseline performance (Assignment 1).
- **Chunk B (SABR calibration):** calibrate SABR parameters (α, ρ, ν) across days and maturities.
- **Chunk C (SABR & Bartlett hedging + competition):** compute Δ^{SABR}, Δ^{Bart}, compute bucketed SSE/MSE/Gain, and export standardized scoreboards.

Competition score. Your competitive score is:

$$\text{Score} := \text{Gain(Bart vs BS)} = 1 - \frac{\text{SSE}(\Delta^{Bart})}{\text{SSE}(\Delta^{BS})},$$

computed *after* the standardized filters and bucketing in Step 6.

Global notation

Trading days and one-day changes. Let t index trading days. For any end-of-day quantity X_t, define

$$\Delta X_t := X_{t+1} - X_t.$$

Midquote. Define the call option midquote V_t (index points) by

$$V_t := \frac{\text{bid}_t + \text{ask}_t}{2} \quad \text{(use best bid/offer if available).}$$

Maturity. Define days-to-expiration and year fraction:

$$D_t := (T - t) \text{ in days}, \qquad \tau_t := \frac{D_t}{365}.$$

(If you use a different day-count convention, apply it everywhere.)

Underlying and forward proxy. Let S_t denote the SPX close and let

$$F_t := S_t e^{(r-q)\tau_t}$$

be the corresponding forward proxy. Define the one-day hedge move as:

$$\Delta S_t^{\text{hedge}} := \begin{cases} S_{t+1} - S_t, & \text{if you set } q = 0, \\ \Delta S_t + S_t\big(e^{q_t \Delta t} - 1\big), & \text{if you include dividends (total return).} \end{cases}$$

All hedge ratios used in the residual definitions below must therefore be interpreted as hedge ratios with respect to the underlying S_t (or, equivalently, with respect to the one-step hedge move $\Delta S_t^{\text{hedge}}$). Whenever a pricing formula is first written in forward variables, the corresponding hedge ratio must be converted back to an S_t-based hedge ratio by the chain rule.

Delta-hedged residual (one-step). Given a hedge delta δ_t, define the hedging residual

$$\varepsilon_t(\delta) := \Delta V_t - \delta_t\, \Delta S_t^{\text{hedge}}.$$

SSE, MSE, Gain. For a set of observations $\mathcal{I}$, define

$$\text{SSE}_{\mathcal{I}}(\delta) := \sum_{t \in \mathcal{I}} \varepsilon_t(\delta)^2, \qquad \text{MSE}_{\mathcal{I}}(\delta) := \frac{1}{|\mathcal{I}|}\text{SSE}_{\mathcal{I}}(\delta).$$

Given two delta hedges $\delta^{(A)}$ and $\delta^{(B)}$, define

$$\text{Gain}(A \text{ vs } B) := 1 - \frac{\text{SSE}(\delta^{(A)})}{\text{SSE}(\delta^{(B)})},$$

where both SSEs are computed over the *same* observation set.

Important: Standardization for competition. You must follow Step 6 filters, bin edges, and output schema exactly.

Tasks

1. **Step 1: Data loading and cleaning (reproducibility)**
   Load the option panel and apply the following preprocessing:

   - Keep calls only: `cp_flag == 'C'`.
   - Convert `date` and `exdate` to `datetime`.
   - Rescale strikes so `strike_price` is in index points (SPX: divide by 1000).
   - Construct the midquote V_t.
   - Remove observations with missing or non-positive σ_{mkt} (implied volatility).

   Required reporting:

   - row counts before/after the call filter,
   - number of unique trading dates and unique expiries,
   - missing rates of $(V_t,\ \sigma_{\text{mkt}},\ \delta_{\text{vendor}})$,
   - final column list used in the project.

2. **Step 2: Contract time series and one-day option changes**
   Construct a contract identifier that tracks the same option over time (e.g. using `symbol`). Within each contract compute:

   $$\Delta V_t = V_{t+1} - V_t,$$

   and keep only observations where ΔV_t is defined.

   Required reporting:

   - summary statistics of V_t and ΔV_t,
   - number of contracts retained after constructing ΔV_t.

3. **Step 3: Underlying SPX merge and one-day underlying changes**
   Obtain the SPX close series $\{S_t\}$ and compute $\Delta S_t = S_{t+1} - S_t$. Merge S_t (and ΔS_t) into the option panel by date.

```python
import pandas as pd

start = "2023-02-01"
end   = "2023-03-02"  # include at least one extra trading day

d1 = pd.Timestamp(start).strftime("%Y%m%d")
d2 = pd.Timestamp(end).strftime("%Y%m%d")

url = f"https://stooq.com/q/d/l/?s=^spx&d1={d1}&d2={d2}&i=d"
df  = pd.read_csv(url, parse_dates=["Date"]).sort_values("Date")

SPX_price_series = df.set_index("Date")["Close"]
```

   Required reporting:

   - merge coverage: % of option observations matched to valid $(S_t, \Delta S_t)$,
   - summary statistics of S_t and ΔS_t,
   - one plot of S_t and one plot of ΔS_t over time.

4. **Step 4 (Chunk A): Baseline BS delta hedging and baseline SSE**
   This step establishes the *baseline* hedge, which is always your **own** BS delta (Assignment 1).

   (a) Baseline delta definition (standardized). Define your baseline delta Δ_t^{BS} by the Black–Scholes(-Merton) formula using *the market implied volatility* $\sigma_{\text{mkt},t}$ for that option and your chosen (r, q) convention. Let $F_t = S_t e^{(r-q)\tau}$ be the forward proxy. The hedge ratio used in this assignment is the *spot* Black–Scholes delta, i.e. the derivative of the option price with respect to the underlying S_t:

   $$\Delta_t^{BS} := e^{-q\tau}\Phi(d_1), \qquad d_1 = \frac{\ln(S_t/K) + (r - q + \frac{1}{2}\sigma_{\text{mkt},t}^2)\tau}{\sigma_{\text{mkt},t}\sqrt{\tau}}.$$

   Equivalently, since $F_t = S_t e^{(r-q)\tau}$,

   $$\Delta_t^{BS} = \frac{\partial C}{\partial S_t} = \frac{\partial B}{\partial F}(F_t, K, \tau, \sigma_{\text{mkt},t})\,\frac{\partial F_t}{\partial S_t} = e^{(r-q)\tau}\frac{\partial B}{\partial F}(F_t, K, \tau, \sigma_{\text{mkt},t}).$$

   Important: Δ^{BS} is the benchmark in all Gains. Vendor deltas (if provided) may be reported, but must *not* replace Δ^{BS} as the benchmark.

(b) Baseline residuals and SSE. Compute residuals

$$\varepsilon_t(\Delta^{BS}) = \Delta V_t - \Delta_t^{BS}\, \Delta S_t^{\text{hedge}},$$

and later (after Step 6 filters) compute

$$\text{SSE}(\Delta^{BS}) = \sum_t \varepsilon_t(\Delta^{BS})^2, \qquad \text{MSE}(\Delta^{BS}) = \frac{1}{n}\text{SSE}(\Delta^{BS}).$$

Required reporting:

- summary statistics of $\varepsilon_t(\Delta^{BS})$,
- baseline $\text{SSE}(\Delta^{BS})$ after standardized filters (Step 6),
- one plot of $\Delta^{BS}(K)$ vs strike for one chosen day and maturity slice,
- a volatility misspecification experiment: hedge with $\Delta_t^{BS}(\sigma)$ for a grid of σ values and discuss the effect on SSE/Gain.

5. **Step 5 (Chunk B): SABR calibration ($\beta = 1$) across dates and maturities**
In this step you calibrate lognormal SABR to implied volatilities.

(a) Model and fixed parameters. We use forward-form SABR with $\theta = 0$ and $\beta = 1$:

$$\begin{aligned} dF(t) &= \sigma(t)F(t)\, dW_1(t), \qquad F(0) = F_t, \\ d\sigma(t) &= \nu\, \sigma(t)\, dW_2(t), \qquad \sigma(0) = \alpha, \\ dW_1(t)\, dW_2(t) &= \rho\, dt, \end{aligned}$$

with parameters $\alpha > 0$, $\nu > 0$, $\rho \in [-1, 1]$. Please see Appendix B for all analytical derivations needed in this assignment regarding the SABR model.

(b) Strike selection (mandatory). For each trading date t and each maturity slice (fixed D days), calibrate the SABR model using the full set of available strikes in that slice. That is, if

$$\mathcal{K}_{t,D} := \{K : \text{an implied volatility quote is available on date } t \text{ for maturity } D\},$$

then the calibration must use all strikes in $\mathcal{K}_{t,D}$. In the standardized implementation used in this assignment, only maturity slices satisfying

$$1 \le D \le 200$$

are considered.

(c) Hagan lognormal approximation for $\beta = 1$. Let F be the forward proxy and τ the year fraction. Define

$$z := \frac{\nu}{\alpha} \ln\left(\frac{F}{K}\right), \qquad x(z) := \ln\left(\frac{\sqrt{1 - 2\rho z + z^2} + z - \rho}{1 - \rho}\right).$$

For $F \neq K$,

$$\sigma_{\text{imp}}^{SABR}(K; F, \tau; \alpha, \rho, \nu) = \alpha \frac{z}{x(z)} \left(1 + \left[\frac{\rho\nu\alpha}{4} + \frac{2 - 3\rho^2}{24}\nu^2\right]\tau\right).$$

For ATM ($F = K$), use the stable limit:

$$\sigma_{\text{imp}}^{SABR}(F; F, \tau; \alpha, \rho, \nu) = \alpha \left(1 + \left[\frac{\rho\nu\alpha}{4} + \frac{2 - 3\rho^2}{24}\nu^2\right]\tau\right).$$

You must implement the ATM branch robustly. Please see Appendix B for all the theory and derivations.

(d) Calibration objective, bounds, and initialization (mandatory). For each slice (t, D), estimate (α, ρ, ν) via nonlinear least squares:

$$(\hat{\alpha}_{t,D}, \hat{\rho}_{t,D}, \hat{\nu}_{t,D}) \in \arg \min_{\alpha>0,\ \nu>0,\ \rho\in[-1,1]} \sum_{K\in\mathcal{K}_{t,D}} \left(\sigma_{\text{mkt}}(K; t, D) - \sigma_{\text{imp}}^{SABR}(K; F_t, \tau; \alpha, \rho, \nu)\right)^2.$$

Please see Appendix B for analytical gradients of the implied volatility and of the Black price with respect to the model parameters, which are key ingredients for gradient-based variants of the calibration objective and for good numerical convergence of the optimization routine.

(e) Diagnostics to store. For each calibrated slice (t, D), store:

$$|\mathcal{K}_{t,D}|, \quad \hat{\alpha}_{t,D},\ \hat{\rho}_{t,D},\ \hat{\nu}_{t,D}, \quad \mathrm{RMSE}^{\mathrm{vol}}_{t,D}.$$

Required reporting:

- at least one plot of "market smile vs SABR fitted smile" for a chosen (t, D),
- time series plots of $\hat{\alpha}_{t,D}$ and $\hat{\nu}_{t,D}$ for at least two maturities,
- a calibration error analysis (MAE/MSE or RMSE) summarized across the (t, D, K) grid.

6. **Step 6 (Chunk C): SABR delta, Bartlett delta, standardized filters/buckets, and Gains** This step defines deltas and the standardized evaluation.

(a) Black call greeks (forward form). Use the Black formula for a call on forward F:

$$B(F, K, \tau, \sigma) = e^{-r\tau} \left(F\Phi(d_1) - K\Phi(d_2) \right), \quad d_1 = \frac{\ln(F/K) + \frac{1}{2}\sigma^2\tau}{\sigma\sqrt{\tau}}, \quad d_2 = d_1 - \sigma\sqrt{\tau}.$$

Then

$$\frac{\partial B}{\partial F} = e^{-r\tau}\Phi(d_1), \qquad \frac{\partial B}{\partial \sigma} = e^{-r\tau} F\sqrt{\tau}\,\varphi(d_1).$$

Since the hedging residuals in this assignment are computed against $\Delta S_t^{\mathrm{hedge}}$, the final hedge ratios must be expressed as derivatives with respect to S_t. Using

$$F_t = S_t e^{(r-q)\tau}, \qquad \frac{\partial F_t}{\partial S_t} = e^{(r-q)\tau},$$

the corresponding S_t-based hedge ratios are obtained by multiplying the forward-based chain-rule expressions by $e^{(r-q)\tau}$.

(b) SABR delta (smile-aware chain rule). Let $\sigma_{\mathrm{imp}} = \sigma_{\mathrm{imp}}^{SABR}(K; F_t, \tau; \hat{\alpha}, \hat{\rho}, \hat{\nu})$. Define first the forward-based smile-aware delta

$$\Delta_t^{SABR,F} := \frac{\partial B}{\partial F}(F_t, K, \tau, \sigma_{\mathrm{imp}}) + \frac{\partial B}{\partial \sigma}(F_t, K, \tau, \sigma_{\mathrm{imp}}) \cdot \frac{\partial \sigma_{\mathrm{imp}}}{\partial F}(F_t, K, \tau).$$

The hedge ratio used in the hedging residuals is the corresponding S_t-based hedge ratio

$$\Delta_t^{SABR} := \frac{\partial C}{\partial S_t} = e^{(r-q)\tau}\Delta_t^{SABR,F}.$$

Equivalently,

$$\Delta_t^{SABR} = e^{(r-q)\tau}\left[\frac{\partial B}{\partial F}(F_t, K, \tau, \sigma_{\mathrm{imp}}) + \frac{\partial B}{\partial \sigma}(F_t, K, \tau, \sigma_{\mathrm{imp}}) \cdot \frac{\partial \sigma_{\mathrm{imp}}}{\partial F}(F_t, K, \tau)\right].$$

Derivative computation: you may compute $\partial\sigma_{\mathrm{imp}}/\partial F$ either analytically from Appendix B or numerically by central differences. If the numerical route is used, take

$$h_F := 10^{-4}F_t, \qquad \frac{\partial \sigma_{\mathrm{imp}}}{\partial F} \approx \frac{\sigma_{\mathrm{imp}}(F_t + h_F) - \sigma_{\mathrm{imp}}(F_t - h_F)}{2h_F}.$$

(c) Bartlett delta (skew adjustment). Bartlett delta adds the SABR "optimal" skew adjustment at the forward level:

$$\Delta_t^{Bart,F} := \frac{\partial B}{\partial F} + \frac{\partial B}{\partial \sigma}\left(\frac{\partial \sigma_{\mathrm{imp}}}{\partial F} + \frac{\partial \sigma_{\mathrm{imp}}}{\partial \alpha}\,\frac{\rho\,\nu}{F_t^{\beta}}\right), \qquad \beta = 1.$$

Thus in this lab,

$$\Delta_t^{Bart,F} = \frac{\partial B}{\partial F} + \frac{\partial B}{\partial \sigma}\left(\frac{\partial \sigma_{\mathrm{imp}}}{\partial F} + \frac{\partial \sigma_{\mathrm{imp}}}{\partial \alpha}\,\frac{\rho\,\nu}{F_t}\right).$$

The hedge ratio used in the residuals is again the corresponding S_t-based hedge ratio:

$$\Delta_t^{Bart} := \frac{\partial C}{\partial S_t} = e^{(r-q)\tau}\Delta_t^{Bart,F},$$

that is,

$$\Delta_t^{Bart} = e^{(r-q)\tau}\left[\frac{\partial B}{\partial F} + \frac{\partial B}{\partial \sigma}\left(\frac{\partial \sigma_{\text{imp}}}{\partial F} + \frac{\partial \sigma_{\text{imp}}}{\partial \alpha}\frac{\rho\,\nu}{F_t}\right)\right].$$

Derivative computation: you may compute $\partial\sigma_{\text{imp}}/\partial\alpha$ either analytically from Appendix B or numerically by central differences. If the numerical route is used, take

$$h_\alpha := 10^{-4}\hat{\alpha}_{t,D}, \qquad \frac{\partial \sigma_{\text{imp}}}{\partial \alpha} \approx \frac{\sigma_{\text{imp}}(\alpha + h_\alpha) - \sigma_{\text{imp}}(\alpha - h_\alpha)}{2h_\alpha}.$$

The appendix B provides analytical formulas for the derivatives of the SABR implied-volatility approximation. Hence, for the computation of $\partial\sigma_{\text{imp}}/\partial F$ and $\partial\sigma_{\text{imp}}/\partial\alpha$, you may use either the corresponding analytical expressions from the appendix or a numerical differentiation scheme. If numerical differentiation is used, the method and step size must be stated explicitly. In all cases, the chosen differentiation procedure must be implemented consistently with the SABR model specification and notation adopted in this assignment.

Concept questions (short answers). Explain the role of the correlation ρ in the skew adjustment. What sign would you expect for ρ in an equity index market with leverage effect? Provide one plot (fixed (t, D)) showing the effect of $\beta \in \{0, 0.5, 1\}$ on $K \mapsto \Delta^{SABR}(K)$ and $K \mapsto \Delta^{Bart}(K)$.

(d) Standardized filters (mandatory). After computing all deltas, filter out observations with:

$$\Delta_t^{BS} \le 0.05, \qquad \Delta_t^{BS} \ge 0.95, \qquad D_t \le 14,$$

and additionally drop rows where SABR calibration is missing (so Δ^{SABR} and Δ^{Bart} are undefined). Report the remaining row count after each filter.

(e) Standardized bucketing (mandatory, fixed bin edges). Create 9 moneyness bins based on Δ^{BS} and 7 maturity bins based on D_t, using the fixed edges.

(f) Hedging residuals and SSE/MSE. Compute residuals

$$\varepsilon_t(\Delta^{BS}) = \Delta V_t - \Delta_t^{BS}\Delta S_t^{\text{hedge}}, \quad \varepsilon_t(\Delta^{SABR}) = \Delta V_t - \Delta_t^{SABR}\Delta S_t^{\text{hedge}}, \quad \varepsilon_t(\Delta^{Bart}) = \Delta V_t - \Delta_t^{Bart}\Delta S_t^{\text{hedge}},$$

and compute total SSE/MSE on the final filtered sample:

$$\text{SSE}(\Delta^{BS}),\ \text{SSE}(\Delta^{SABR}),\ \text{SSE}(\Delta^{Bart}), \qquad \text{MSE}(\Delta^{BS}),\ \text{MSE}(\Delta^{SABR}),\ \text{MSE}(\Delta^{Bart}).$$

Also compute bucketed SSE/MSE for each bucket (i, j).

(g) Gains (total and bucketed). Compute total Gains:

$$\text{Gain}(SABR \text{ vs } BS) = 1 - \frac{\text{SSE}(\Delta^{SABR})}{\text{SSE}(\Delta^{BS})}, \qquad \text{Gain}(Bart \text{ vs } BS) = 1 - \frac{\text{SSE}(\Delta^{Bart})}{\text{SSE}(\Delta^{BS})}.$$

Compute relative Gains (Bartlett vs SABR), **always using your own BS baseline**:

$$\text{RelGain}(Bart \text{ vs } SABR) = 1 - \frac{\text{SSE}(\Delta^{Bart})}{\text{SSE}(\Delta^{SABR})}.$$

Compute the same quantities per bucket (i, j) by replacing SSE with $\text{SSE}_{i,j}$.

Required plots:

- Smile fit: $\sigma_{\text{mkt}}(K)$ vs $\sigma_{\text{imp}}^{SABR}(K)$ for one chosen (t, D).
- Delta comparison: $K \mapsto \Delta^{BS}(K)$, $K \mapsto \Delta^{SABR}(K)$, $K \mapsto \Delta^{Bart}(K)$ for one chosen (t, D).
- Heatmap of $\text{Gain}_{i,j}(Bart \text{ vs } BS)$.
- Heatmap of $\text{RelGain}_{i,j}(Bart \text{ vs } SABR)$.

7. **Step 7: Standardized outputs for competition (CSV + PDF)**
To participate in the ranking, you must export the following **exact** files:

(a) `CalibrationMetrics.csv` (one row per calibrated slice (t, D)) with columns:

```
date, D_days, tau, n_strikes, alpha, rho, nu, vol_rmse
```

(b) `HedgingScoreboard.csv` in **tidy long format** (one row per bucket (`bin1`, `bin2`)) plus one optional TOTAL row. Required columns:

```
section, bin1, bin2, n_obs,
sse_bs, sse_sabr, sse_bartlett,
mse_bs, mse_sabr, mse_bartlett,
gain_sabr_vs_bs, gain_bart_vs_bs, relgain_bart_vs_sabr
```

Formatting rules (mandatory):

- Bucket rows must have `section="BUCKET"` and integer `bin1`, `bin2`.
- The TOTAL row (if included) must have `section="TOTAL"` and `bin1="TOTAL"`, `bin2="TOTAL"`.
- Gains must be computed using your own BS baseline Δ^{BS} (not vendor delta).

A Assignment 1 - The Black-Scholes heat equation and Exotic option pricing

The aim of this appendix is to present the canonical financial model namely the Black-Scholes Merton model, and to use it as a backbone to introduce the different numerical methods for PDE solvers. Let us first remind ourselfs the Black-Scholes PDE and how this is derived.

Derivation of the Black-Scholes PDE

This paragraph is intended to provide a rigorous derivation of the so-called Black-Scholes partial differential equation. Let $(W_t)_{t\geq 0}$ be a Brownian motion and $S := (S_t)_{t\geq 0}$ the asset price. This model assumes the following dynamics under the so-called historical (observed) probability measure $\mathbb{P}$

$$\frac{dS_t}{S_t} = \mu dt + \sigma dW_t, \quad S_0 > 0 \tag{7}$$

with $\mu \in \mathbb{R}$ is called the drift and $\sigma > 0$ the instanteneous volatility. Let $V(t,s)$ be the value of a European call option. The question we are interested in here is the following; assuming that the dynamics of the underlying asset is given by the above stochastic differential equation, what is the value today ($t = 0$) of a European option with payoff $f(S_T)$ at maturity $T > 0$? For clarity we shall denote V_t the value at time $t \in [0,T]$ of such financial derivative. The first step is to obtain a probabilistic representation for the option price. Under the absence absence of arbitrage there exists a probability $\mathbb{Q}$ under which we can write

$$V_0 = D_{0,T}\mathbb{E}_{\mathbb{Q}}\left[f(S)\right] \tag{8}$$

where $(D_{0,t})_{t\geq 0}$ represents the discount factor proces satisfying the (stochastic) differential equation $dD_{0,t} = -r_t D_{0,t}dt, \quad D_{0,0} = 1$. We will simply assume a constant interest rate hence we just have a exponential growth ode which gives us an exponential discounting. An application of Itô's lemma yields that the option price satisfies the stochastic differential equation

$$dV_t = \left(\mu S_t \partial_S V_t + \partial_t V_t + \frac{\sigma^2}{2} S_t^2 \partial_{SS}^2 V_t\right) dt + \sigma S_t \partial_S V_t dW_t \tag{9}$$

at any time t between inception and maturity with appropriate boundary conditions. Consider now a portfolio Π consisting at time t of a long position in the option V and long position in Δ_t shares S, i.e $\Pi_t = V_t + \Delta_t S_t$. On a small time interval $[t, t+dt]$ the profit and loss of such a portfolio is $d\Pi_t = dV_t + \Delta_t dS_t$. Using the equation (9) we obtain that

$$d\Pi_t = \{\mu\left(\Delta_t + \partial_S V_t\right) S_t + \partial_t V_t + \frac{\sigma^2}{2} S_t^2 \partial_{SS}^2 V_t\} dt + \left(\Delta + \partial_S V_t\right)\sigma S_t dW_t \tag{10}$$

This expression makes it clear that the only way to eliminate the risk—solely present in the form of the Brownian perturbations- is to set $\Delta_t = -\partial_S V_t$. This is called Delta-hedging and you have already worked with this in the first assignment. now since we assume the absence of arbitrage, the returns of the portfolio Π_t over the period $[t, t+dt]$ are necessarily equal to the risk-free rate $r_t \equiv r$. Otherwise (assume the returns are higher than the risk-free one), it is possible to construct an arbitrage, for instance by borrowing money at time t to buy the portfolio, then invest it at rate r_t and sell it at time $t + dt$. Hence our assumption on the absence of arbitrage implies that $d\Pi_t = r\Pi_t dt$ and hence from equation (10)

$$\partial_t V_t + rS\partial_S V_t + \frac{\sigma^2}{2} S^2 \partial_{SS}^2 V_t = rV_t \tag{11}$$

This equation (11) is called the Black-Scholes PDE associated with the boundary conditions given by the payoff $V_T = f(S_T)$.

Before trying to solve (numerically) a partial differential equation, it may sound sensible to simplify it. Recall that the BS PDE above is a parabolic PDE with boundary conditions $V_T(S)$ (for instance a European call option with maturity $T > 0$ and strike $K > 0$, we have $V_T(S) = (S_T - K)^+$). Define $\tau := T - t$ and the function $g_\tau(S) := V_t(S)$, then $\partial_t V_t(S) = -\partial_\tau g_\tau(S)$ and hence

$$-\partial_\tau g_\tau + rS\partial_S g_\tau + \frac{\sigma^2}{2} S^2 \partial_{SS}^2 g_\tau = rg_\tau \tag{12}$$

with boundary conditions $g_0(S)$. Define now the function f by $f_\tau(S) := e^{r\tau} g_\tau(S)$. Show that

1. $$-\partial_\tau f_\tau + rS\partial_S f_\tau + \frac{\sigma^2}{2} S^2 \partial_{SS}^2 f_\tau = 0 \tag{13}$$

   with boundary conditions $f_0(S)$.

2. Consider now a further trnasformation $x := \log(S)$ and the function $\psi_\tau(x) := f_\tau(S)$. Show that this transformation gives us the following PDE

$$-\partial_\tau \psi_\tau + \left(r - \frac{\sigma^2}{2}\right)\partial_x \psi_\tau + \frac{\sigma^2}{2}\partial^2_{xx}\psi_\tau = 0 \tag{14}$$

with boundary condition $\psi_0(x)$.

- Finally define the function ϕ_τ via $\psi_\tau(x) =: \phi_\tau(x)e^{\alpha x + \beta\tau}$. Show that equation (14) becomes

$$\partial_\tau \phi_\tau(x) = \frac{\sigma^2}{2}\partial^2_{xx}\phi_\tau(x) \tag{15}$$

for all real number x with (Dirichlet) boundary condition $\phi_0(x) = e^{-\alpha x}\psi_0(x)$. This as you have already seen in class is the so called *heat-equation*.

Options on Extrema

We know that Vanilla optiosn with payoff $C = \phi(S_T)$ can be priced as

$$e^{-rT}\mathbb{E}\left[\phi(S_T)\right] = e^{-rT}\int_0^\infty \phi(y)\phi_{S_T}(y)dy \tag{16}$$

where $\phi_{S_T}(y)$ is the probability density function of S_T which satisfies

$$\mathbb{Q}\left(S_T \le y\right) = \int_0^y \phi_{S_T}(\nu)d\nu, \quad y > 0 \tag{17}$$

Recall that typically we have that $\phi(x) = (x-K)^+$ for the European call option with strike K and $\phi(x) = \mathbb{1}_{[K,\infty)}(x)$ for the binary call option. On the other hand exotic options also called path-dependent options whose payoff depends on the whole path of the underlying price process. For example the payoff of an option on extrema take the form

$$C := \phi(M_0^T, S_T) \tag{18}$$

where $M_0^T = \max_{t\in[0,T]} S_t$ is the maximum of $(S_t)_{t\in\mathbb{R}_+}$ over the time interval $[0,T]$. In such situations option price at time $t=0$ can be expressed as

$$e^{-rT}\mathbb{E}\left[\phi(M_0^T, S_T)\right] = e^{-rT}\int_0^\infty\int_0^\infty \phi(x,y)\phi_{M_0^T,S_T}(x,y)dxdy \tag{19}$$

where $\phi_{M_0^T,S_T}$ is the joint probability density function of (M_0^T, S_T) which satisfies

$$\mathbb{Q}\left(M_0^T \le x, S_T \le y\right) = \int_0^x\int_0^y \phi_{M_0^T,S_T}(u,v)dudv \tag{20}$$

Using the joint probability density function of the shifted Brownian motion $\tilde{W}_T = W_T + \mu T$ and the

$$\hat{X}_0^T = \max_{t\in[0,T]} \tilde{W} = \max_{t\in[0,T]}(W_t + \mu t) \tag{21}$$

We have that the joint probability density function of $\phi_{\hat{X}_0^T,\tilde{W}_T}$ of the drifted Brownian motion and its maximum is given by

$$\phi_{\hat{X}_0^T,\tilde{W}_T}(\alpha,b) = \mathbb{1}_{\{\alpha\ge\max(b,0)\}}\frac{1}{T}\sqrt{\frac{2}{\pi T}}\left(2\alpha - b\right)e^{\mu b - (2\alpha-b)^2/(2T) - \mu^2 T/2} \tag{22}$$

Using this we are able to price any exotic option with payoff $\phi(\tilde{W}_T, \hat{X_0}^T)$ as

$$c_0 = e^{-rT}\int_{-\infty}^\infty\int_{y\vee 0}^\infty \phi(x,y)d\mathbb{Q}\left(\tilde{W}_T \le x, \hat{X_0}^T \le y\right) \tag{23}$$

Notice that so far we have not assumed any dynamics on the underlying asset. We will now assume that S_t follows a GBM. In order to price barrier options by the above probabilistic method we will use the probability density function of the maximum

$$M_0^T = \max_{t\in[0,T]} S_t \tag{24}$$

of the GBM over a given time interval and the joint pdf $\phi_{M_0^T,S_T}(u,\nu)$.

Lemma A.1. *An exotic option with integrable claim payoff of the form*

$$C = \phi(M_0^T, S_T) = \phi(\max_{t\in[0,T]} S_t, S_T) \tag{25}$$

can be priced at time $t = 0$ as

$$\begin{aligned} e^{-rT}\mathbb{E}\left[C\right] &= \frac{e^{-rT}}{T^{3/2}}\sqrt{\frac{2}{\pi}}\int_0^\infty \int_y^\infty \phi(S_0 e^{\sigma y}, S_0 e^{\sigma x})(2x-y)e^{-\mu^2 T/2+\mu y-(2x-y)^2/(2T)}dxdy \\ &+ \frac{e^{-rT}}{T^{3/2}}\sqrt{\frac{2}{\pi}}\int_{-\infty}^0 \int_0^\infty \phi(S_0 e^{\sigma y}, S_0 e^{\sigma x})e^{-\mu^2 T/2+\mu y-(2x-y)^2/(2T)}dxdy \end{aligned}$$

B Appendix: The SABR model and analytic formulas

In this appendix we collect the analytical formulas underlying the SABR specification used in Assignment 2. Throughout, we work on a fixed calibration slice (t, D), freeze the time-t forward level, and write

$$f := F_t, \qquad \tau := \tau_t, \qquad K > 0,$$

so that the SABR-implied volatility is viewed as a deterministic function of $(f, K, \tau; \alpha, \rho, \nu, \beta, \theta)$. In the assignment itself we use the special case

$$\theta = 0, \qquad \beta = 1,$$

but it is useful to state the general formulas first and then specialize them.

A.1. SABR dynamics

The SABR model is given by

$$dF(u) = \sigma(u)\big(F(u)+\theta\big)^{\beta} dW_1(u), \qquad F(0) = f, \tag{26}$$

$$d\sigma(u) = \nu\,\sigma(u)\,dW_2(u), \qquad \sigma(0) = \alpha, \tag{27}$$

$$d\langle W_1, W_2\rangle_u = \rho\,du, \qquad -1 \le \rho \le 1, \tag{28}$$

where:

- $F(u)$ is the forward process,
- $\sigma(u)$ is the instantaneous volatility process,
- α is the initial volatility level,
- $\nu > 0$ is the volatility-of-volatility parameter,
- ρ is the instantaneous correlation between the forward shock and the volatility shock,
- $\beta \in [0, 1]$ controls the elasticity of the forward diffusion,
- θ is a shift parameter.

For the assignment we work with the *lognormal SABR model*, that is

$$\theta = 0, \qquad \beta = 1,$$

so that the forward dynamics reduce to

$$dF(u) = \sigma(u)F(u)\,dW_1(u), \qquad d\sigma(u) = \nu\sigma(u)\,dW_2(u).$$

A.2. Hagan-type short-maturity expansion of the implied volatility

Let

$$x := \log\left(\frac{f}{K}\right)$$

denote log-moneyness. For fixed (f, K), the SABR implied Black volatility admits the short-maturity expansion

$$\sigma_{\text{imp}}^{\text{SABR}}(K; f, \tau; \alpha, \rho, \nu, \beta) = I_B^0(x)\big(1 + \tau I_H^1(x)\big) + O(\tau^2). \tag{29}$$

The leading term $I_B^0(x)$ is the zero-order implied volatility and $I_H^1(x)$ is the first-order time correction.

First-order time correction. The first-order correction is

$$I_H^1(x) = \frac{\alpha^2 (Kf)^{\beta-1}(1-\beta)^2}{24} + \frac{\alpha\beta\rho\nu(Kf)^{\frac{\beta}{2}-\frac{1}{2}}}{4} + \frac{\nu^2(2-3\rho^2)}{24}. \tag{30}$$

Corrected zero-order term. Introduce the auxiliary quantity

$$z := \begin{cases} \dfrac{\nu}{\alpha}\,\dfrac{f^{1-\beta}-K^{1-\beta}}{1-\beta}, & \beta \neq 1, \\[2mm] \dfrac{\nu}{\alpha}\log\!\left(\dfrac{f}{K}\right), & \beta = 1, \end{cases} \tag{31}$$

and define

$$\xi := \sqrt{1-2\rho z + z^2}, \qquad \chi(z) := \log\!\left(\frac{\xi + z - \rho}{1-\rho}\right). \tag{32}$$

Then the corrected zero-order implied volatility can be written as

$$I_B^0(x) = \begin{cases} \dfrac{\nu x}{\chi(z)}, & x \neq 0,\ \nu \neq 0, \\[2mm] \alpha K^{\beta-1}, & x = 0, \\[2mm] \alpha\,\dfrac{(1-\beta)x}{f^{1-\beta}-K^{1-\beta}}, & \nu = 0. \end{cases} \tag{33}$$

Hence the full first-order approximation is

$$\sigma_{\text{imp}}^{\text{SABR}} = I_B^0(x)\big(1 + \tau I_H^1(x)\big). \tag{34}$$

Remark on consistency. For $\beta < 1$, the representation (33) is the corrected zero-order term that is consistent as $\beta \to 1$. In particular,

$$\lim_{\beta\to 1} \frac{\nu}{\alpha}\,\frac{f^{1-\beta}-K^{1-\beta}}{1-\beta} = \frac{\nu}{\alpha}\log\!\left(\frac{f}{K}\right),$$

so the general formula reduces continuously to the $\beta = 1$ case.

A.3. Special cases of the zero-order term I_B^0

For later differentiation it is convenient to isolate the main cases explicitly.

Case 1: ATM case $x = 0$. When $f = K$, equivalently $x = 0$, one has

$$I_B^0 = \alpha K^{\beta-1}. \tag{35}$$

Case 2: zero vol-of-vol $\nu = 0$. When $\nu = 0$,

$$I_B^0 = \frac{x\alpha(1-\beta)}{f^{1-\beta}-K^{1-\beta}}. \tag{36}$$

Case 3: lognormal SABR $\beta = 1$. When $\beta = 1$,

$$z = \frac{\nu x}{\alpha}, \qquad \xi = \sqrt{1-2\rho z + z^2}, \qquad I_B^0 = \frac{\nu x}{\chi(z)}. \tag{37}$$

Case 4: CEV-type SABR $\beta < 1$. When $\beta < 1$,

$$z = \frac{\nu\big(f^{1-\beta}-K^{1-\beta}\big)}{\alpha(1-\beta)}, \qquad \xi = \sqrt{1-2\rho z + z^2}, \qquad I_B^0 = \frac{\nu x}{\chi(z)}. \tag{38}$$

A.4. Derivatives of the first-order correction I_H^1

From (30) one obtains the derivatives

$$\frac{\partial I_H^1}{\partial \alpha} = \frac{\alpha (Kf)^{\beta-1}(1-\beta)^2}{12} + \frac{\beta\rho\nu (Kf)^{\frac{\beta}{2}-\frac{1}{2}}}{4}, \tag{39}$$

$$\frac{\partial I_H^1}{\partial \beta} = \frac{\alpha^2 (Kf)^{\beta-1}(1-\beta)^2 \log(Kf)}{24} + \frac{\alpha^2 (Kf)^{\beta-1}(2\beta-2)}{24} + \frac{\alpha\beta\rho\nu (Kf)^{\frac{\beta}{2}-\frac{1}{2}} \log(Kf)}{8} + \frac{\alpha\rho\nu (Kf)^{\frac{\beta}{2}-\frac{1}{2}}}{4}, \tag{40}$$

$$\frac{\partial I_H^1}{\partial \nu} = \frac{\alpha\beta\rho (Kf)^{\frac{\beta}{2}-\frac{1}{2}}}{4} + \frac{\nu(2-3\rho^2)}{12}, \tag{41}$$

$$\frac{\partial I_H^1}{\partial \rho} = \frac{\alpha\beta\nu (Kf)^{\frac{\beta}{2}-\frac{1}{2}}}{4} - \frac{\rho\nu^2}{4}, \tag{42}$$

$$\frac{\partial I_H^1}{\partial f} = \frac{\alpha^2 K^{\beta-1}(\beta-1)^3}{24} f^{\beta-2} + \frac{\alpha\beta\rho\nu \left(\frac{\beta}{2}-\frac{1}{2}\right) K^{\frac{\beta}{2}-\frac{1}{2}}}{4} f^{\frac{\beta}{2}-\frac{3}{2}}, \tag{43}$$

$$\frac{\partial I_H^1}{\partial K} = \frac{\alpha^2 (Kf)^{\beta-1}(1-\beta)^2(\beta-1)}{24K} + \frac{\alpha\beta\rho\nu (Kf)^{\frac{\beta}{2}-\frac{1}{2}} \left(\frac{\beta}{2}-\frac{1}{2}\right)}{4K}. \tag{44}$$

The second derivatives that are needed for the higher-order smile Greeks are

$$\frac{\partial^2 I_H^1}{\partial f^2} = \frac{\alpha^2(\beta-2)(Kf)^{\beta-1}(\beta-1)^3}{24f^2} + \frac{\alpha\beta\rho\nu (Kf)^{\frac{\beta}{2}-\frac{1}{2}} \left(\frac{\beta}{2}-\frac{1}{2}\right)\left(\frac{\beta}{2}-\frac{3}{2}\right)}{8f^2}, \tag{45}$$

$$\frac{\partial^2 I_H^1}{\partial f\,\partial \alpha} = \frac{\alpha (Kf)^{\beta-1}(1-\beta)^2(\beta-1)}{12f} + \frac{\beta\rho\nu (Kf)^{\frac{\beta}{2}-\frac{1}{2}} \left(\frac{\beta}{2}-\frac{1}{2}\right)}{4f}, \tag{46}$$

$$\frac{\partial^2 I_H^1}{\partial \alpha^2} = \frac{(Kf)^{\beta-1}(1-\beta)^2}{12}. \tag{47}$$

A.5. Differential calculus for the zero-order term I_B^0

The cleanest way to differentiate I_B^0 is to treat it as the composite map

$$I_B^0(x) = \frac{\nu x}{\chi(z)}, \qquad \chi(z) = \log\left(\frac{\sqrt{1-2\rho z + z^2} + z - \rho}{1-\rho}\right).$$

Let

$$\xi(z) := \sqrt{1-2\rho z + z^2}.$$

Then

$$\chi'(z) = \frac{1}{\xi(z)}, \qquad \chi''(z) = -\frac{z-\rho}{\xi(z)^3}. \tag{48}$$

Moreover, for $\beta \neq 1$,

$$z = \frac{\nu}{\alpha}\frac{f^{1-\beta} - K^{1-\beta}}{1-\beta}, \tag{49}$$

and the required derivatives are

$$\frac{\partial z}{\partial f} = \frac{\nu}{\alpha} f^{-\beta}, \qquad \frac{\partial z}{\partial K} = -\frac{\nu}{\alpha} K^{-\beta}, \qquad \frac{\partial z}{\partial \alpha} = -\frac{z}{\alpha}, \qquad \frac{\partial z}{\partial \nu} = \frac{z}{\nu}, \tag{50}$$

$$\frac{\partial^2 z}{\partial f^2} = -\frac{\beta\nu}{\alpha} f^{-\beta-1}, \qquad \frac{\partial^2 z}{\partial f\,\partial \alpha} = -\frac{\nu}{\alpha^2} f^{-\beta}, \qquad \frac{\partial^2 z}{\partial \alpha^2} = \frac{2z}{\alpha^2}. \tag{51}$$

For $\beta = 1$, these formulas reduce to

$$z = \frac{\nu}{\alpha}\log\left(\frac{f}{K}\right), \qquad \frac{\partial z}{\partial f} = \frac{\nu}{\alpha f}, \qquad \frac{\partial z}{\partial K} = -\frac{\nu}{\alpha K}, \qquad \frac{\partial^2 z}{\partial f^2} = -\frac{\nu}{\alpha f^2}.$$

Compact first-derivative rule. For any parameter $p \in \{f, K, \alpha, \nu, \beta, \rho\}$,

$$\frac{\partial I_B^0}{\partial p} = \frac{\partial(\nu x)}{\partial p}\frac{1}{\chi(z)} - \frac{\nu x}{\chi(z)^2}\,\chi'(z)\,\frac{\partial z}{\partial p}. \tag{52}$$

Since $\chi'(z) = 1/\xi(z)$, this becomes

$$\frac{\partial I_B^0}{\partial p} = \frac{\partial(\nu x)}{\partial p}\frac{1}{\chi(z)} - \frac{\nu x}{\xi(z)\chi(z)^2}\,\frac{\partial z}{\partial p}. \tag{53}$$

Compact second-derivative rule. For $p, q \in \{f, \alpha\}$,

$$\begin{aligned}\frac{\partial^2 I_B^0}{\partial p\,\partial q} &= \frac{\partial^2(\nu x)}{\partial p\,\partial q}\frac{1}{\chi} - \frac{\partial(\nu x)}{\partial p}\frac{\chi'}{\chi^2}z_q - \frac{\partial(\nu x)}{\partial q}\frac{\chi'}{\chi^2}z_p \\ &\quad - (\nu x)\left[\frac{\chi''}{\chi^2} - 2\frac{(\chi')^2}{\chi^3}\right] z_p z_q - (\nu x)\frac{\chi'}{\chi^2}z_{pq},\end{aligned} \tag{54}$$

where, for brevity, $z_p = \partial_p z$, $z_{pq} = \partial^2_{pq} z$, $\chi = \chi(z)$, $\chi' = \chi'(z)$, $\chi'' = \chi''(z)$.

A.5.1. Explicit derivatives in the ATM case $x = 0$

From (35) one obtains

$$\frac{\partial I_B^0}{\partial \alpha} = K^{\beta-1}, \qquad \frac{\partial I_B^0}{\partial \beta} = \alpha K^{\beta-1}\log K, \qquad \frac{\partial I_B^0}{\partial \nu} = 0, \qquad \frac{\partial I_B^0}{\partial \rho} = 0, \tag{55}$$

$$\frac{\partial I_B^0}{\partial f} = 0, \qquad \frac{\partial I_B^0}{\partial K} = \alpha(\beta-1)K^{\beta-2}, \qquad \frac{\partial^2 I_B^0}{\partial f^2} = 0, \qquad \frac{\partial^2 I_B^0}{\partial \alpha^2} = 0, \qquad \frac{\partial^2 I_B^0}{\partial f\,\partial \alpha} = 0. \tag{56}$$

A.5.2. Explicit derivatives in the case $\nu = 0$

From (36),

$$\frac{\partial I_B^0}{\partial \alpha} = \frac{x(1-\beta)}{f^{1-\beta} - K^{1-\beta}}, \tag{57}$$

$$\frac{\partial I_B^0}{\partial \beta} = \frac{\alpha x\left(K^{1-\beta} - f^{1-\beta} + (\beta-1)\left(K^{1-\beta}\log K - f^{1-\beta}\log f\right)\right)}{\left(K^{1-\beta} - f^{1-\beta}\right)^2}, \tag{58}$$

$$\frac{\partial I_B^0}{\partial \nu} = 0, \qquad \frac{\partial I_B^0}{\partial \rho} = 0, \tag{59}$$

$$\frac{\partial I_B^0}{\partial f} = \frac{\alpha(\beta-1)\left(f\left(K^{1-\beta} - f^{1-\beta}\right) - f^{2-\beta}x(\beta-1)\right)}{f^2\left(K^{1-\beta} - f^{1-\beta}\right)^2}, \tag{60}$$

$$\frac{\partial I_B^0}{\partial K} = \frac{\alpha(\beta-1)\left(-K\left(K^{1-\beta} - f^{1-\beta}\right) + K^{2-\beta}x(\beta-1)\right)}{K^2\left(K^{1-\beta} - f^{1-\beta}\right)^2}, \tag{61}$$

and the second derivatives needed later are

$$\frac{\partial^2 I_B^0}{\partial f^2} = \frac{\alpha(\beta-1)\Big(-f^4\left(K^{1-\beta} - f^{1-\beta}\right)^2 + f^{5-\beta}\left(K^{1-\beta} - f^{1-\beta}\right)(\beta-1)\big(x(\beta-1) + x - 2\big) + 2f^{6-2\beta}x(\beta-1)^2\Big)}{f^6\left(K^{1-\beta} - f^{1-\beta}\right)^3}, \tag{62}$$

$$\frac{\partial^2 I_B^0}{\partial \alpha^2} = 0, \tag{63}$$

$$\frac{\partial^2 I_B^0}{\partial f\,\partial \alpha} = -\frac{f^{1-\beta}x(1-\beta)^2}{f\left(-K^{1-\beta} + f^{1-\beta}\right)^2} + \frac{1-\beta}{f\left(-K^{1-\beta} + f^{1-\beta}\right)}. \tag{64}$$

A.5.3. Explicit derivatives in the lognormal case $\beta = 1$

In the case $\beta = 1$, from (37),

$$I_B^0 = \frac{\nu x}{\chi(z)}, \qquad z = \frac{\nu x}{\alpha}, \qquad \xi = \sqrt{1 - 2\rho z + z^2}, \qquad \chi(z) = \log\left(\frac{\xi + z - \rho}{1 - \rho}\right).$$

Applying (53) and (54) gives

$$\frac{\partial I_B^0}{\partial \alpha} = \frac{\nu x z}{\alpha \xi \chi(z)^2}, \tag{65}$$

$$\frac{\partial I_B^0}{\partial \beta} = 0, \tag{66}$$

$$\frac{\partial I_B^0}{\partial \nu} = \frac{x\big(\alpha \xi \chi(z) - \nu x\big)}{\alpha \xi \chi(z)^2}, \tag{67}$$

$$\frac{\partial I_B^0}{\partial \rho} = \frac{\nu x\left((\rho - 1)(z + \xi) + (-\rho + z + \xi)\xi\right)}{(\rho - 1)(-\rho + z + \xi)\xi\, \chi(z)^2}, \tag{68}$$

$$\frac{\partial I_B^0}{\partial f} = \frac{\nu\big(\alpha \xi \chi(z) - \nu x\big)}{\alpha f\, \xi\, \chi(z)^2}, \tag{69}$$

$$\frac{\partial I_B^0}{\partial K} = \frac{\nu\big(-\xi \alpha \chi(z) + \nu x\big)}{K\, \xi\, \alpha\, \chi(z)^2}. \tag{70}$$

The second derivatives required later are

$$\frac{\partial^2 I_B^0}{\partial f^2} = \frac{\nu\Big(-\xi^3 \alpha^2 \chi(z)^2 + \xi^2 \alpha \nu x \chi(z) - 2\xi^2 \alpha \nu \chi(z) + 2\xi \nu^2 x - \rho \nu^2 x \chi(z) + \nu^2 x z \chi(z)\Big)}{\xi^3 \alpha^2 f^2 \chi(z)^3}, \tag{71}$$

$$\frac{\partial^2 I_B^0}{\partial \alpha^2} = \frac{\nu x z\Big(-2\xi^2 \chi(z) + 2\xi z - \rho z \chi(z) + z^2 \chi(z)\Big)}{\xi^3 \alpha^2 \chi(z)^3}, \tag{72}$$

$$\frac{\partial^2 I_B^0}{\partial f\, \partial \alpha} = \frac{\nu\Big(\xi^2 \alpha z \chi(z) + \xi^2 \nu x \chi(z) - 2\xi \nu x z + \rho \nu x z \chi(z) - \nu x z^2 \chi(z)\Big)}{\xi^3 \alpha^2 f \chi(z)^3}. \tag{73}$$

A.5.4. Explicit derivatives in the case $\beta < 1$

In the general case $\beta < 1$, from (38),

$$z = \frac{\nu(f^{1-\beta} - K^{1-\beta})}{\alpha(1 - \beta)}, \qquad I_B^0 = \frac{\nu x}{\chi(z)}.$$

Using the chain rule (53), one obtains

$$\frac{\partial I_B^0}{\partial \alpha} = \frac{\nu x z}{\alpha \xi \chi(z)^2}, \tag{74}$$

$$\frac{\partial I_B^0}{\partial \beta} = \frac{K^{-\beta} f^{-\beta} \nu x\left(K f^{\beta} \nu \log K + K^{\beta} \alpha f^{\beta} z - K^{\beta} f \nu \log f\right)}{\xi \alpha (\beta - 1) \chi(z)^2}, \tag{75}$$

$$\frac{\partial I_B^0}{\partial \nu} = \frac{x\left(\xi \alpha (\beta - 1) \chi(z) - \nu\left(K^{1-\beta} - f^{1-\beta}\right)\right)}{\xi \alpha (\beta - 1) \chi(z)^2}, \tag{76}$$

$$\frac{\partial I_B^0}{\partial \rho} = \frac{\nu x\left(\xi(\xi - \rho + z) + (\xi + z)(\rho - 1)\right)}{\xi(\rho - 1)(\xi - \rho + z)\chi(z)^2}, \tag{77}$$

$$\frac{\partial I_B^0}{\partial f} = \frac{f^{-\beta} \nu\left(\xi \alpha f^{\beta} \chi(z) - f \nu x\right)}{\xi \alpha f\, \chi(z)^2}, \tag{78}$$

$$\frac{\partial I_B^0}{\partial K} = \frac{K^{-\beta} \nu\left(K \nu x - K^{\beta} \xi \alpha \chi(z)\right)}{K\, \xi\, \alpha\, \chi(z)^2}. \tag{79}$$

The second derivatives needed below are

$$\frac{\partial^2 I_B^0}{\partial f^2} = \frac{\nu}{\xi^3\alpha^2 f^3(\xi-\rho+z)\chi(z)^3}\Bigg[-\xi^3\alpha^2 f(\xi-\rho+z)\chi(z)^2 - 2\xi^2\alpha f^{2-\beta}\nu(\xi-\rho+z)\chi(z)$$
$$+\xi f^{3-2\beta}\nu^2 x(\xi-\rho+z)\chi(z) + 2\xi f^{3-2\beta}\nu^2 x(\xi-\rho+z)$$
$$+ f\nu x\Big(\xi^3\alpha\beta f^{1-\beta} + \xi^2\Big(\alpha f^{1-\beta}(-\rho(\beta-1)-\rho+z) - \nu\big(f^{1-\beta}(-K^{1-\beta}+f^{1-\beta}) + f^{2-2\beta}\big)\Big)$$
$$+ f^{2-2\beta}\nu(\rho-z)^2\Big)\chi(z)\Bigg], \tag{80}$$

$$\frac{\partial^2 I_B^0}{\partial \alpha^2} = \frac{\nu x z\Big(-2\xi^2\chi(z) + 2\xi z - \rho z\chi(z) + z^2\chi(z)\Big)}{\xi^3\alpha^2\chi(z)^3}, \tag{81}$$

$$\frac{\partial^2 I_B^0}{\partial f\,\partial \alpha} = \frac{f^{-\beta}\nu\Big(\xi^2\alpha f^{\beta} z\chi(z) + \xi^2 f\nu x\chi(z) - 2\xi f\nu x z + f\rho\nu x z\chi(z) - f\nu x z^2\chi(z)\Big)}{\xi^3\alpha^2 f\chi(z)^3}, \tag{82}$$

A.6. Jacobian and Hessian of the SABR implied volatility

Recall that

$$\sigma_{\text{imp}}^{\text{SABR}} = I_B^0(x)\big(1+\tau I_H^1(x)\big).$$

Hence, for any parameter p,

$$\frac{\partial \sigma_{\text{imp}}^{\text{SABR}}}{\partial p} = \frac{\partial I_B^0}{\partial p}\big(1+\tau I_H^1\big) + I_B^0\,\tau\,\frac{\partial I_H^1}{\partial p}. \tag{83}$$

In particular,

$$\frac{\partial \sigma_{\text{imp}}^{\text{SABR}}}{\partial \alpha} = \frac{\partial I_B^0}{\partial \alpha}\big(1+\tau I_H^1\big) + I_B^0\,\tau\,\frac{\partial I_H^1}{\partial \alpha}, \tag{84}$$

$$\frac{\partial \sigma_{\text{imp}}^{\text{SABR}}}{\partial \beta} = \frac{\partial I_B^0}{\partial \beta}\big(1+\tau I_H^1\big) + I_B^0\,\tau\,\frac{\partial I_H^1}{\partial \beta}, \tag{85}$$

$$\frac{\partial \sigma_{\text{imp}}^{\text{SABR}}}{\partial \nu} = \frac{\partial I_B^0}{\partial \nu}\big(1+\tau I_H^1\big) + I_B^0\,\tau\,\frac{\partial I_H^1}{\partial \nu}, \tag{86}$$

$$\frac{\partial \sigma_{\text{imp}}^{\text{SABR}}}{\partial \rho} = \frac{\partial I_B^0}{\partial \rho}\big(1+\tau I_H^1\big) + I_B^0\,\tau\,\frac{\partial I_H^1}{\partial \rho}, \tag{87}$$

$$\frac{\partial \sigma_{\text{imp}}^{\text{SABR}}}{\partial K} = \frac{\partial I_B^0}{\partial K}\big(1+\tau I_H^1\big) + I_B^0\,\tau\,\frac{\partial I_H^1}{\partial K}, \tag{88}$$

$$\frac{\partial \sigma_{\text{imp}}^{\text{SABR}}}{\partial f} = \frac{\partial I_B^0}{\partial f}\big(1+\tau I_H^1\big) + I_B^0\,\tau\,\frac{\partial I_H^1}{\partial f}. \tag{89}$$

The second derivatives required for higher-order smile Greeks are

$$\frac{\partial^2 \sigma_{\text{imp}}^{\text{SABR}}}{\partial f^2} = \frac{\partial^2 I_B^0}{\partial f^2} + \tau\left(\frac{\partial^2 I_B^0}{\partial f^2} I_H^1 + I_B^0\frac{\partial^2 I_H^1}{\partial f^2} + 2\frac{\partial I_B^0}{\partial f}\frac{\partial I_H^1}{\partial f}\right), \tag{90}$$

$$\frac{\partial^2 \sigma_{\text{imp}}^{\text{SABR}}}{\partial \alpha\,\partial f} = \frac{\partial^2 I_B^0}{\partial \alpha\,\partial f}\big(1+\tau I_H^1\big) + \tau\frac{\partial I_B^0}{\partial \alpha}\frac{\partial I_H^1}{\partial f} + \tau I_B^0\frac{\partial^2 I_H^1}{\partial \alpha\,\partial f} + \tau\frac{\partial I_H^1}{\partial \alpha}\frac{\partial I_B^0}{\partial f}, \tag{91}$$

$$\frac{\partial^2 \sigma_{\text{imp}}^{\text{SABR}}}{\partial \alpha^2} = \frac{\partial^2 I_B^0}{\partial \alpha^2} + \tau\left(\frac{\partial^2 I_B^0}{\partial \alpha^2} I_H^1 + I_B^0\frac{\partial^2 I_H^1}{\partial \alpha^2} + 2\frac{\partial I_B^0}{\partial \alpha}\frac{\partial I_H^1}{\partial \alpha}\right). \tag{92}$$

A.7. Black formula and smile-aware Greeks

Let

$$B(f,K,\tau,\Sigma) = e^{-r\tau}\big(f\Phi(d_1) - K\Phi(d_2)\big), \tag{93}$$

with

$$d_1 = \frac{\log(f/K) + \frac{1}{2}\Sigma^2\tau}{\Sigma\sqrt{\tau}}, \qquad d_2 = d_1 - \Sigma\sqrt{\tau}. \tag{94}$$

Then

$$\frac{\partial B}{\partial f} = e^{-r\tau}\Phi(d_1), \tag{95}$$

$$\frac{\partial B}{\partial \Sigma} = e^{-r\tau} f\sqrt{\tau}\,\varphi(d_1), \tag{96}$$

$$\frac{\partial^2 B}{\partial f^2} = e^{-r\tau}\frac{\varphi(d_1)}{f\Sigma\sqrt{\tau}}, \tag{97}$$

$$\frac{\partial^2 B}{\partial f\,\partial \Sigma} = -e^{-r\tau}\frac{\varphi(d_1)d_2}{\Sigma}, \tag{98}$$

$$\frac{\partial^2 B}{\partial \Sigma^2} = e^{-r\tau} f\sqrt{\tau}\,\varphi(d_1)\frac{d_1 d_2}{\Sigma}. \tag{99}$$

For notational convenience, write

$$\Delta_B := \frac{\partial B}{\partial f}, \qquad \mathcal{V}_B := \frac{\partial B}{\partial \Sigma}, \qquad \Gamma_B := \frac{\partial^2 B}{\partial f^2}, \qquad \mathrm{Vanna}_B := \frac{\partial^2 B}{\partial f\,\partial \Sigma}, \qquad \mathrm{Volga}_B := \frac{\partial^2 B}{\partial \Sigma^2}.$$

A.8. SABR delta and Bartlett delta

Set

$$\Sigma(f,K,\tau) := \sigma_{\text{imp}}^{\text{SABR}}(K; f, \tau; \alpha, \rho, \nu, \beta).$$

Then the smile-aware SABR delta is

$$\Delta^{SABR} = \frac{\partial B}{\partial f} + \frac{\partial B}{\partial \Sigma}\frac{\partial \Sigma}{\partial f} = \Delta_B + \mathcal{V}_B \Sigma_f. \tag{100}$$

The Bartlett delta augments this with the skew-covariation correction:

$$\Delta^{Bart} = \frac{\partial B}{\partial f} + \frac{\partial B}{\partial \Sigma}\left(\frac{\partial \Sigma}{\partial f} + \frac{\partial \Sigma}{\partial \alpha}\frac{\rho\nu}{f^\beta}\right) = \Delta_B + \mathcal{V}_B\left(\Sigma_f + \Sigma_\alpha \frac{\rho\nu}{f^\beta}\right). \tag{101}$$

For puts, put–call parity gives

$$\Delta_{put}^{Bart} = \Delta_{call}^{Bart} - 1. \tag{102}$$

Lognormal specialization $\beta = 1$. In the assignment, $\beta = 1$, and therefore

$$\Delta^{Bart} = \Delta_B + \mathcal{V}_B\left(\Sigma_f + \Sigma_\alpha \frac{\rho\nu}{f}\right). \tag{103}$$

A.9. Higher-order smile Greeks

Define

$$A(f) := \Sigma_f + \Sigma_\alpha \frac{\rho\nu}{f^\beta}.$$

Then (101) becomes

$$\Delta^{Bart} = \Delta_B + \mathcal{V}_B A.$$

Differentiating once more with respect to f yields the Bartlett gamma

$$\Gamma^{Bart} = \Gamma_B + 2\,\mathrm{Vanna}_B\, A + \mathrm{Volga}_B\, A^2 + \mathcal{V}_B\, A_f, \tag{104}$$

where

$$A_f = \Sigma_{ff} + \Sigma_{\alpha f}\frac{\rho\nu}{f^\beta} - \Sigma_\alpha \frac{\beta\rho\nu}{f^{\beta+1}}. \tag{105}$$

Hence

$$\begin{aligned}\Gamma^{Bart} = \Gamma_B &+ 2\,\mathrm{Vanna}_B\left(\Sigma_f + \Sigma_\alpha \frac{\rho\nu}{f^\beta}\right) + \mathrm{Volga}_B\left(\Sigma_f + \Sigma_\alpha \frac{\rho\nu}{f^\beta}\right)^2 \\ &+ \mathcal{V}_B\left(\Sigma_{ff} + \Sigma_{\alpha f}\frac{\rho\nu}{f^\beta} - \Sigma_\alpha \frac{\beta\rho\nu}{f^{\beta+1}}\right).\end{aligned} \tag{106}$$

Following the usual SABR Greek terminology, one may also define the α-vega, rega, sega, vanna and volga by

$$\mathcal{V} := \frac{\partial C}{\partial \alpha} = \mathcal{V}_B \left(\Sigma_\alpha + \Sigma_f \frac{\rho f^\beta}{\nu} \right), \tag{107}$$

$$\mathcal{R} := \frac{\partial C}{\partial \rho} = \mathcal{V}_B \, \Sigma_\rho, \tag{108}$$

$$\mathcal{S} := \frac{\partial C}{\partial \nu} = \mathcal{V}_B \, \Sigma_\nu, \tag{109}$$

$$\begin{aligned} \text{Vanna} := \frac{\partial \mathcal{V}}{\partial f} &= \left(\text{Vanna}_B + \text{Volga}_B \left(\Sigma_f + \Sigma_\alpha \frac{\rho \nu}{f^\beta} \right) \right) \left(\Sigma_\alpha + \Sigma_f \frac{\rho f^\beta}{\nu} \right) \\ &+ \mathcal{V}_B \left(\Sigma_{f\alpha} + \Sigma_{ff} \frac{\rho f^\beta}{\nu} + \Sigma_f \frac{\beta \rho f^{\beta - 1}}{\nu} \right), \end{aligned} \tag{110}$$

$$\text{Volga} := \frac{\partial \mathcal{V}}{\partial \alpha} = \text{Volga}_B \left(\Sigma_\alpha + \Sigma_f \frac{\rho f^\beta}{\nu} \right)^2 + \mathcal{V}_B \left(\Sigma_{\alpha\alpha} + \Sigma_{f\alpha} \frac{\rho f^\beta}{\nu} \right). \tag{111}$$

A.10. Simplifications for the assignment: $\theta = 0$ and $\beta = 1$

In Assignment 2, the SABR model is used with

$$\theta = 0, \qquad \beta = 1.$$

In this case,

$$I_H^1 = \frac{\rho \nu \alpha}{4} + \frac{2 - 3\rho^2}{24} \nu^2,$$

hence I_H^1 is independent of f and K, and therefore

$$\frac{\partial I_H^1}{\partial f} = \frac{\partial I_H^1}{\partial K} = \frac{\partial^2 I_H^1}{\partial f^2} = \frac{\partial^2 I_H^1}{\partial f \, \partial \alpha} = 0.$$

Consequently,

$$\Sigma_f = (1 + \tau I_H^1) \frac{\partial I_B^0}{\partial f}, \tag{112}$$

$$\Sigma_\alpha = (1 + \tau I_H^1) \frac{\partial I_B^0}{\partial \alpha} + \tau I_B^0 \left(\frac{\rho \nu}{4} \right), \tag{113}$$

$$\Sigma_{ff} = (1 + \tau I_H^1) \frac{\partial^2 I_B^0}{\partial f^2}, \tag{114}$$

$$\Sigma_{f\alpha} = (1 + \tau I_H^1) \frac{\partial^2 I_B^0}{\partial f \, \partial \alpha} + \tau \frac{\partial I_B^0}{\partial f} \left(\frac{\rho \nu}{4} \right), \tag{115}$$

$$\Sigma_{\alpha\alpha} = (1 + \tau I_H^1) \frac{\partial^2 I_B^0}{\partial \alpha^2} + 2\tau \frac{\partial I_B^0}{\partial \alpha} \left(\frac{\rho \nu}{4} \right). \tag{116}$$

Therefore the formulas that are actually needed in the assignment are:

- the lognormal SABR implied volatility

  $$\Sigma = \sigma_{\text{imp}}^{\text{SABR}}(K; f, \tau; \alpha, \rho, \nu) = \frac{\nu x}{\chi(z)} \left(1 + \tau \left[\frac{\rho \nu \alpha}{4} + \frac{2 - 3\rho^2}{24} \nu^2 \right] \right), \qquad z = \frac{\nu x}{\alpha},$$

  with its ATM limit

  $$\Sigma(f; f, \tau; \alpha, \rho, \nu) = \alpha \left(1 + \tau \left[\frac{\rho \nu \alpha}{4} + \frac{2 - 3\rho^2}{24} \nu^2 \right] \right),$$

- the SABR delta

  $$\Delta^{SABR} = \Delta_B + \mathcal{V}_B \Sigma_f,$$

- and the Bartlett delta

  $$\Delta^{Bart} = \Delta_B + \mathcal{V}_B \left(\Sigma_f + \Sigma_\alpha \frac{\rho \nu}{f} \right).$$

Remark on implementation. The appendix provides the analytical derivatives for reference and for theoretical interpretation. In the assignment itself, the official grading procedure uses the standardized numerical differentiation rules stated in Step 6, namely central differences in f and in α. This ensures that all submissions are evaluated under exactly the same approximation scheme.

A.11. Connection with variance-optimal hedging

The theoretical motivation for Bartlett's delta is that in a dynamic implied-volatility setting the variance-optimal hedge takes the form

$$\theta_t^{VO} = \Delta_t^{BS} + \mathcal{V}_t^{BS} \frac{d\langle \Sigma_t, S_t \rangle}{d\langle S_t, S_t \rangle}. \tag{117}$$

In the lognormal SABR model, this variance-optimal correction coincides with Bartlett's delta. Therefore, within SABR, the additional term

$$\mathcal{V}_B \, \Sigma_\alpha \, \frac{\rho \nu}{f^\beta}$$

is not an ad hoc modification of delta, but the precise correction induced by the covariation between the underlying and the stochastic implied-volatility surface.

References

[1] Keller-Ressel, M. Bartlett's delta revisited: Variance-optimal hedging in the lognormal sabr and in the rough bergomi model. Papers 2207.13573, arXiv.org, Jul 2022.